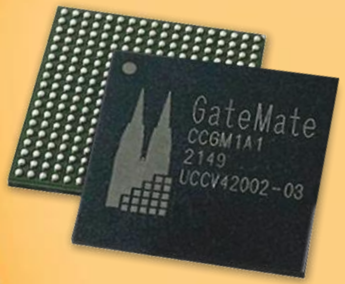
GateMate^{TM} FPGA

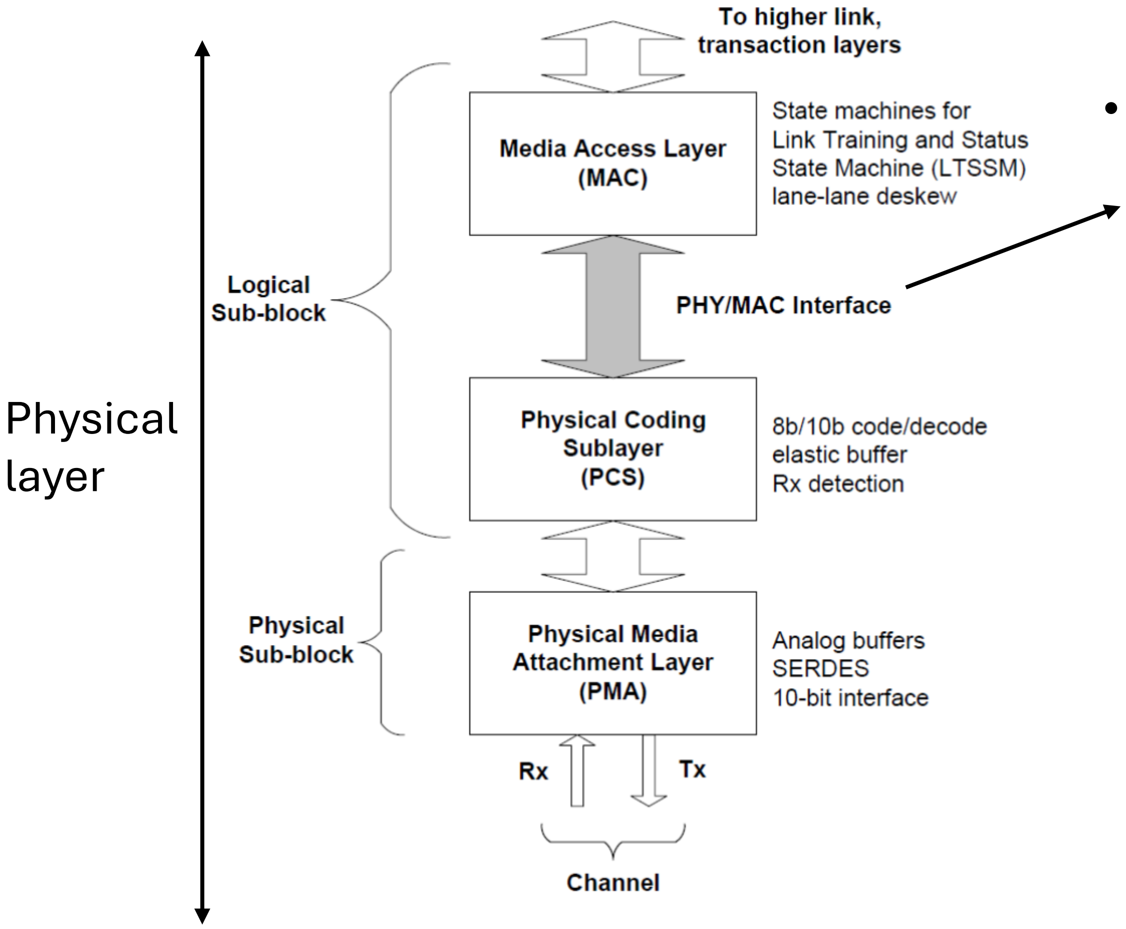
MAC layer



Vinh Hai Luong, B.Sc.
Research&Development
Cologne Chip AG

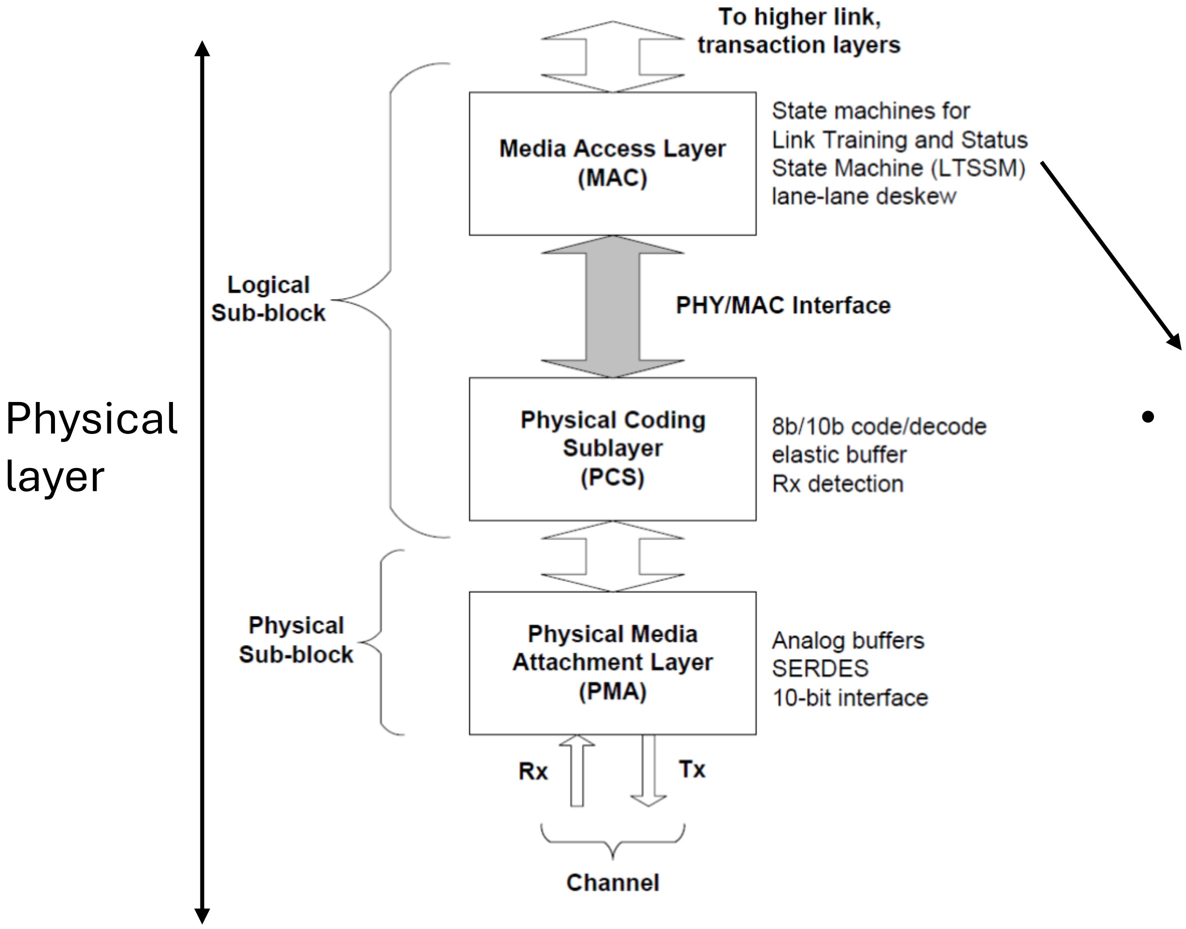
July 2025

PIPE Overview



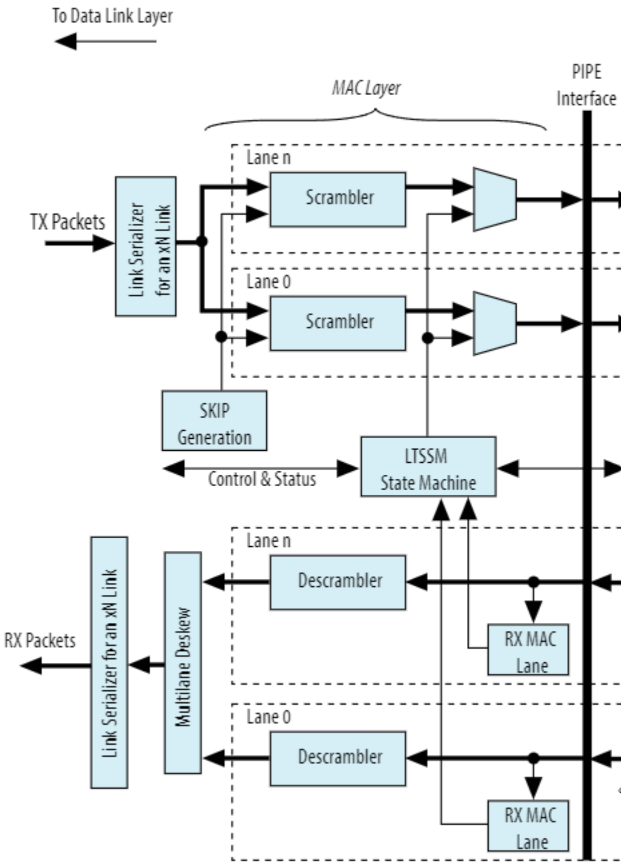
- The **PHY Interface for the PCI Express Architecture (PIPE)** is part of the physical layer and serves as an interface between the Media Access Layer (MAC) and the Physical Coding Sublayer (PCS).

MAC Overview



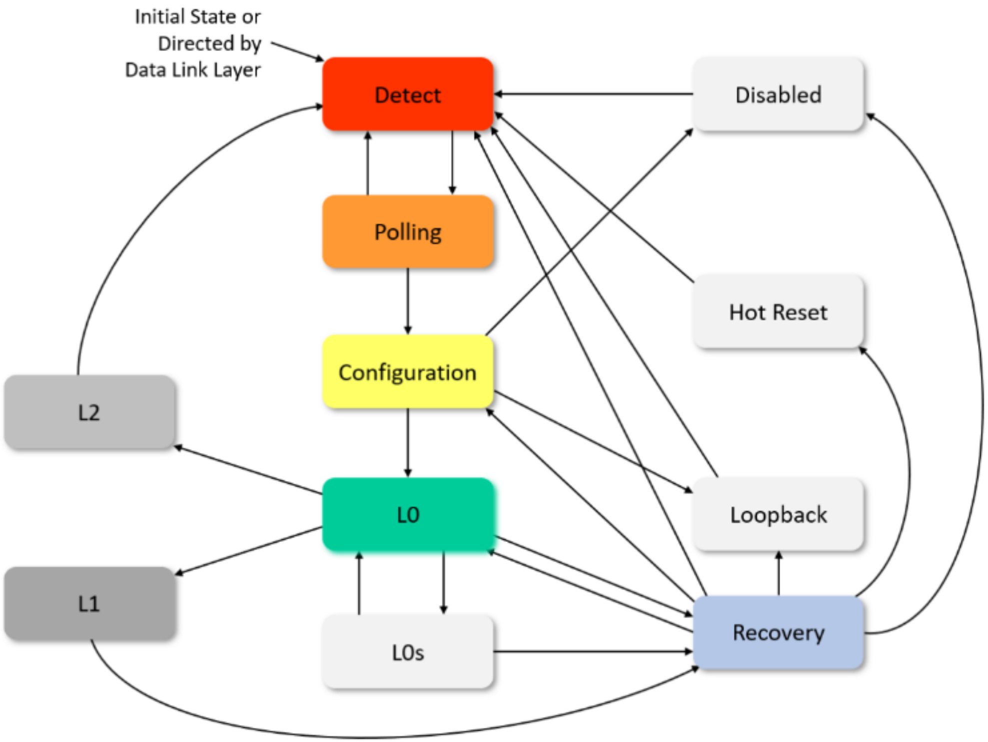
- The **Media Access Layer (MAC)** is part of the physical layer. It lies in between the **PIPE Interface** and the **Data Link Layer (DLL).**

MAC Overview



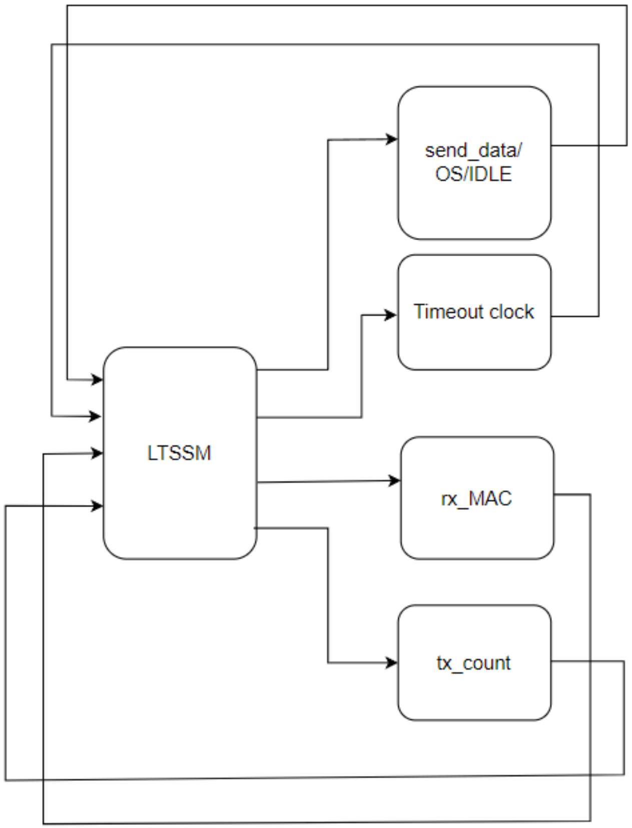
- Scrambler/Descrambler to decode/encode data.
- SKP Generator to insert SKP Ordered Set at an interval between 1180 and 1538 Symbol Times, compensating different bit rates of two ports.
- LTSSM to control Link training and get the Link up.

LTSSM


Initial State or
Directed by
Data Link Layer
Detect
Polling
Configuration
L2
L0
L1
L0s
Disabled
Hot Reset
Loopback
Recovery

LTSSM Block Diagram

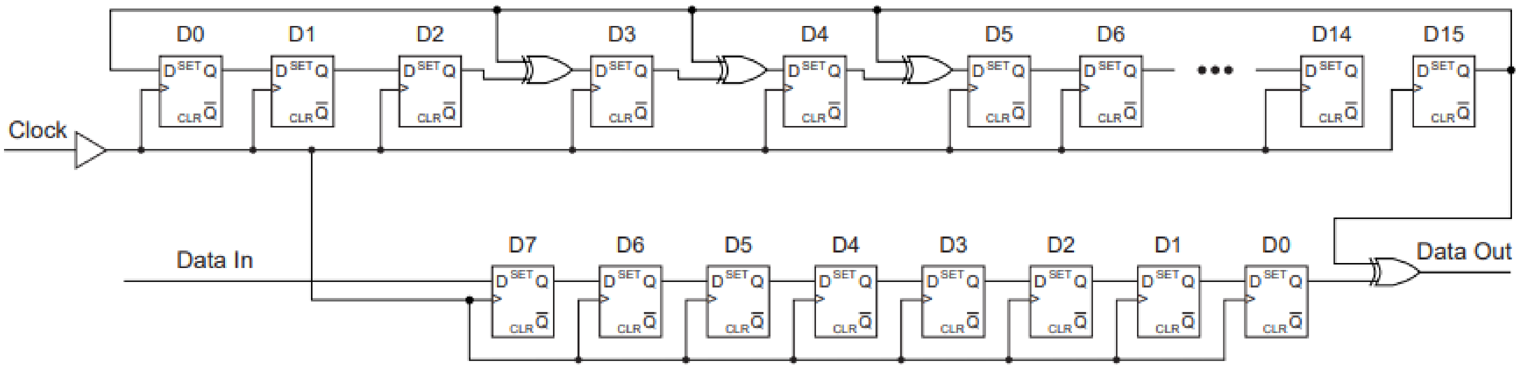
LTSSM Logic



- Send_data/OS/IDLE: Assemble the transmitted data, e.g. TS1 and TS2 Ordered Sets.
- Timeout clock: Trigger timeout clock at each state according to the Specs.
- Rx_MAC:
  - Detect Link/Lane number from received Ordered Sets.
  - Check if received Ordered Sets are valid based on previously determined Link/Lane number.
  - Count the number of consecutive correctly received Ordered Sets and feedback the count flag to the LTSSM.
  - Detect and remove SKP Ordered Sets from received sequence.
- Tx_count: Count the number of transmitted Ordered Set and feedback the count flag to LTSSM

Scrambling/Descrambling

- Scrambling and Descrambling implemented with LFSR.
- Scrambling and Descrambling is done for each symbol.
- LFSR polynomial: $X^{16}+X^5+X^4+X^3+1$
- Rules:
    - COM initializes LFSR.
    - Initialized value is 16’hFFFF.
    - LFSR advances eight shifts for each symbol except for SKP.
    - All data characters except for those in Training Sequence are scrambled.
    - All special characters (K symbols) are not scrambled.
    - Scrambling can only be disabled at the end of Configuration.



Polarity Inversion

- During training sequence, Symbols 6-15 of TS1 and TS2 will be considered to determine if Polarity inversion is necessary or not.
- TS1 Symbols 6-15 received will be D21.5 as opposed to the expected D10.2.
- Symbols 6-15 of the TS2 ordered set will be D26.5 as opposed to the expected D5.2.
- This will be signaled to the Rx of SERDES using o_RxPolarity output port of MAC layer.
- Transmitter does not have to consider polarity inversion.

References

- PHY Interface for the PCI Express* Architecture (PIPE) Specs
- Simon Southwell’s PCIe Primer
- PCI Express* Solutions User Guide

Cologne Chip

Contact

Luong Vinh Hai, B.Sc.
Cologne Chip AG

Eintrachtstr. 113
50668 Köln